

Customer Cartography:

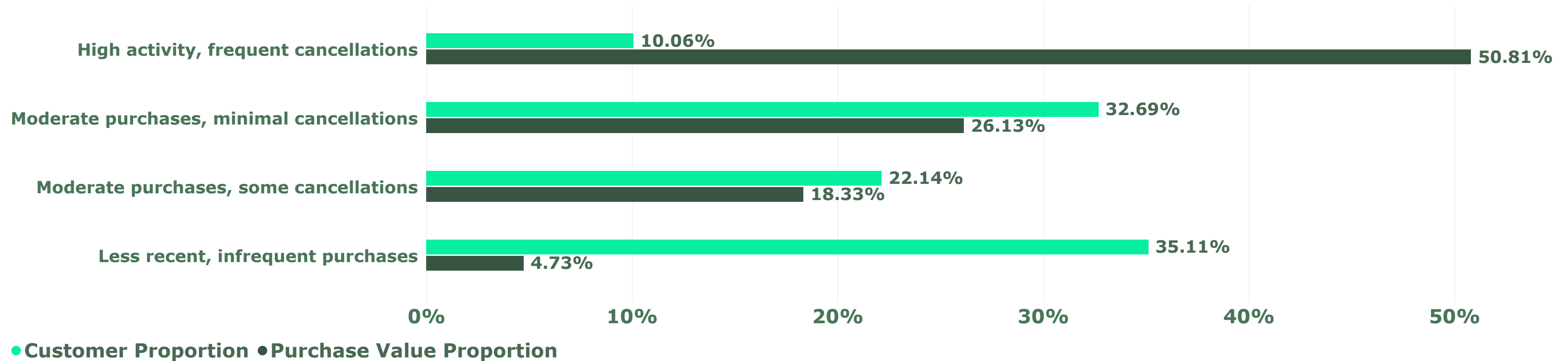
Customer Segmentation by Purchase and Cancellation Activity

Total Customers Analyzed
4,332

Total Purchase Invoices
18,399

Total Purchase Value
£8,514,105.55

Customer Proportion vs. Purchase Value Proportion by Segment



Key Takeaways

- The 436 high activity customers comprise 10.1% of analyzed customers but account for 50.8% of purchase value.
- Less recent, infrequent purchasers form the largest segment, with 1,521 customers representing 35.1% of the customer base but only 4.7% of purchase value.

Source: UCI Online Retail dataset | Dec. 1, 2010–Dec. 9, 2011 | Scope: 4,332 retained customers
Purchase value is in GBP (£) and before cancellations. Cancellation activity is recorded separately.

Segment Profiles:

Comparing Purchase and Cancellation Activity

Customer Behavior by Segment

Segment	Median Days Since Purchase	Median Purchase Invoices	Median Purchase Value (£)	Customers with Cancellations (%)
High activity, frequent cancellations	13	11	£4,611.36	100.00%
Moderate purchases, minimal cancellations	37	3	£912.60	0.14%
Moderate purchases, some cancellations	36	4	£1,186.35	100.00%
Less recent, infrequent purchases	142	1	£226.78	7.43%

Key Takeaways

- The moderate purchasing groups have similar median recency at 37 and 36 days. Cancellation participation differs sharply, at 0.14% for one group and 100% for the other.
- High activity customers made purchases more frequently, with a median of 11 purchase invoices compared with 1 for less recent, infrequent purchasers. Their median total purchase value was also higher at £4,611.36 compared with £226.78.

Recency is measured as of Dec. 10, 2011. Purchase values are in GBP (£) before cancellations. Cancellation participation refers to customers that have at least one cancellation. Data bars are scaled separately by column.

Evidence to Execution:

Targeted Actions and Their Evaluation

Opportunity ▲	Evidence	Targeted Action	Evaluation
Increase purchase value among high activity customers.	The high activity group represents 10.1% of analyzed customers but accounts for 50.8% of purchase value.	Randomly assign high activity customers to two groups. One receives reorder reminders, while the comparison group continues to receive established communications.	Assess whether targeted reminders increase purchase value per assigned customer compared with current communications over the same predefined period. Use purchase frequency, cancellation activity, and campaign costs as supporting measures.
Strengthen understanding of seasonal purchasing	Less recent, infrequent purchasers had a median of one purchase invoice and 142 days since their last purchase. Only 7.43% had cancellation activity.	Examine purchase timing and product types for seasonal patterns. Gather additional transaction data to assess whether these customers return during later holiday or occasion-based periods.	Compare purchasing across equivalent seasonal periods and measure how many customers return, accounting for their available follow-up time.
Explore patterns associated with cancellations	The moderate purchasing groups have similar recency, but only 0.14% of customers in one group had cancellation activity, compared with 100% in the other.	Investigate product mix and order histories for common patterns that may help explain the difference in cancellation activity.	Determine whether product or ordering patterns provide insight beyond the cancellation measures already used to define the groups.

These proposals are based on historical patterns, not proven effects. Evaluation requires new transaction and campaign data. Cancellation investigation also requires reason and order-level details.

Methods and Workflow

The UCI Online Retail transactions were cleaned and organized in MySQL to derive customer purchasing and cancellation features. In Python, these features were transformed and standardized, reduced using PCA, and used to create and evaluate customer segments. The finalized customer data and segment assignments were imported into Power BI to visualize the findings and communicate proposed business actions.

Stage	Approach	Reasoning
1. Organize records	Loaded staging data and built customer, invoice, product, and invoice line tables. Reviewed descriptions and relationships.	Established consistent structure while preserving source records.
2. Define eligibility	Reviewed IDs, prices, special codes, repeated records, and cancellations. Created separate purchase and cancellation views.	Focused on identified customers and eligible merchandise. Retained repeated records because they were not confirmed errors.
3. Derive features	Aggregated purchases and cancellations separately for 4,334 purchasing customers. Derived 12 features with recency referenced to Dec. 10, 2011.	Summarized behavior at the customer level to support segmentation and targeted actions.
4. Review clusters	Reviewed initial KMeans profiles. Excluded two customers with extreme purchases and nearly offsetting cancellations, then refitted on 4,332 customers.	Limited exceptional records' influence without declaring them errors.
5. Transform features	Compared remaining small groups using KMeans and hierarchical clustering. Retained all remaining customers, applied log1p except to recency, and standardized all 12 features.	Reduced skew and the influence of large values without further exclusions.
6. Reduce dimensions	Refit PCA and retained three components explaining 87.16% of standardized variance.	Combined overlapping information into fewer components while preserving most of the differences in customer behavior.
7. Select segments	Compared silhouette scores, sizes, and profiles across cluster counts. Selected four clusters, although two scored highest on silhouette.	Four clusters revealed more detail in purchasing behavior and highlighted cancellation differences between otherwise similar groups.

Two customers were excluded from modeling. The raw records are preserved in the staging table. Model checks and sensitivity results are presented on the Validation page.

Validation:

Evaluating Segment Quality and Consistency

Assessment ▲	Findings	Interpretation
Choosing the number of segments	Two clusters had the highest silhouette score at 0.469. Four clusters scored 0.358 and were selected for their more detailed customer profiles.	Four segments highlighted purchasing and cancellation differences, accepting lower statistical separation.
Changing the random starting point	Customer groupings matched across all six tested random seeds.	The selected solution was consistent across the tested starting points.
Changing how repeated records were treated	Keeping one occurrence of each exact repeated record left 4,322 of 4,332 customers, or 99.77%, in corresponding segments.	The main findings changed very little under this alternative treatment of repeated records.

Median Standardized Feature Values by Segment

Cluster 0 - Moderate purchases, minimal cancellations | Cluster 1 - High activity, frequent cancellations
Cluster 2 - Less recent, infrequent purchases | Cluster 3 - Moderate purchases, some cancellations

Cluster ▲	Recency Days	Purchase Frequency	Purchase Line Count	Distinct Products	Purchase Quantity	Purchase Value	Average Order Value	Cancellation Frequency	Cancellation Line Count	Distinct Cancelled Products	Cancelled Quantity	Cancellation Value
0	-0.56	0.07	0.33	0.39	0.23	0.19	0.26	-0.63	-0.61	-0.61	-0.59	-0.66
1	-0.80	1.68	1.35	1.19	1.40	1.49	0.56	2.22	2.02	1.98	1.81	1.79
2	0.49	-0.95	-0.88	-0.80	-0.85	-0.92	-0.64	-0.63	-0.61	-0.61	-0.59	-0.66
3	-0.57	0.39	0.38	0.37	0.39	0.40	0.20	0.59	0.71	0.76	0.71	0.96

Darker green indicates higher standardized values. Lower recency means more recent purchasing.
These checks support consistency under the conditions tested. They do not establish future stability or prove that proposed business actions will be effective.